

High-Speed Active Quench and Reset Circuit for SPAD in a Standard 65 nm CMOS Technology

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Abstract—A compact high-speed active quench and reset (AQR) circuit integrated with a p^+/n -well single-photon avalanche diode (SPAD) is designed and fabricated in a standard 65 nm CMOS technology. The post-layout simulations showed that the quenching time for this AQR circuit is only 0.1 ns, and the smallest dead time is 3.35 ns which corresponds a maximum count rate of ~ 300 Mcps. The measurements showed that the SPAD pixel achieved a dark count rate of 21 kHz, a peak photon detection probability of 23.8% at a 420 nm wavelength and a timing jitter of 139 ps (using a 405 nm pulsed laser) when the excess voltage was 0.5 V. Also, due to the short quenching time, negligible afterpulsing was observed during the measurements.

Index Terms—Single-photon avalanche diode (SPAD), afterpulsing (AP), dark count rate (DCR), timing jitter, photon detection probability (PDP), active quench and reset (AQR).

I. INTRODUCTION

WITH the advancements in silicon (Si) manufacturing technologies, Si-based photodetectors and light emitting devices are being actively researched (see, for example [1], [2].) One important device is the single-photon avalanche diode (SPAD) especially when it is designed and fabricated in standard complementary metal-oxide-semiconductor (CMOS) technologies. CMOS SPADs have become key photodetectors in many applications such as biomedical imaging [3]–[6], light detection and ranging [7] and communication [8]. This is because they can be fabricated at a lower cost compared to customized processes like high-voltage or CMOS imaging sensor technologies [4] and can be easily integrated with other CMOS front-end and signal processing circuits to achieve strong detector performance at the system level. To ensure the Geiger mode operation, a quench circuit is needed to quench the avalanching process when it is triggered. Also, a reset circuit is required to reset the SPAD back to its initial biasing condition to detect the next photon. One simple method is the passive quench and

reset (PQR) circuit that uses a large resistor connected to either the cathode or anode of the SPAD. The PQR circuits are simple and can be easily implemented in a small layout area in standard CMOS processes. However, the long reset time due to the large quench resistor limits their applications in high-speed detection systems. To address this issue, active quench and reset (AQR) circuits are used in many SPAD pixel designs.

A fast AQR circuit using 0.35 μm CMOS technology was reported in [9]. This fast front-end circuitry consisting of 18 transistors achieved a short active quenching time (T_Q) of ~ 1 ns and a short reset time of 3 ns when the SPAD was biased at 6 V excess voltage (V_{ex}). Due to the short T_Q , the SPAD achieved a low afterpulsing (AP) of 1.3% with 20 ns hold-off time. Researchers in [10] reported a SPAD integrated with a cascaded quenching circuit that was fabricated in a 0.35 μm CMOS technology. The active T_Q was only 0.48 ns, and the dead time (T_D) could be adjusted from 9.5 to 17 ns. The AP was only 0.9% with its highest V_{ex} (6.6 V) and longest T_D (17 ns). An AQR circuit in a 0.18 μm CMOS process in [11] used a positive feedback loop to facilitate the quenching process. The simulation and measurements showed that the T_Q was 0.7 ns with a minimum T_D of 5 ns (corresponding a maximum count rate (CR) of 200 Mcps). The AP was 0.75% with a 4 ns hold-off time.

In this work, a compact high-speed AQR circuit is designed and fabricated using a 65 nm standard CMOS technology. To investigate the performance of the AQR circuit, it is integrated with a p^+/n -well SPAD. The results show that the AQR circuit achieves a fast quenching time, a short dead time and a small layout area because of the advanced technology node.

II. PIXEL DESIGN

A. SPAD Structure

The cross-sectional view of the p^+/n -well SPAD is shown in Fig. 1(a). The diameter of the active area was 10 μm and it had a silicide blocking layer to improve the light absorption. To prevent premature edge breakdown (PEB), an octagonal shape and a p-well guard ring were used. The uniformity of the electric field distribution was verified by the TCAD simulation shown in Fig. 1(b). While the anode terminal had a direct contact to the p^+ region, the presence of the p-well guard ring requires the cathode to be connected through the deep n-well. The p^+ region was extended outside of the active area and into the p-well to push the shallow trench isolation (STI) away from the depletion region of the SPAD to reduce the dark counts caused by the defects at the interface of the STI.

B. AQR Circuit Design

The AQR circuit is shown in Fig. 2 and a sample of the post-layout simulation results are presented in Fig. 3. A SPAD circuit model similar to [12] was used in the simulation. From the I-V characteristics, the avalanching resistance was

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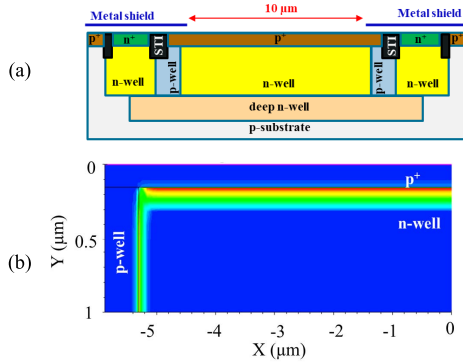


Fig. 1. (a) Cross-sectional view of the p^+/n -well SPAD and (b) the corresponding TCAD simulation.

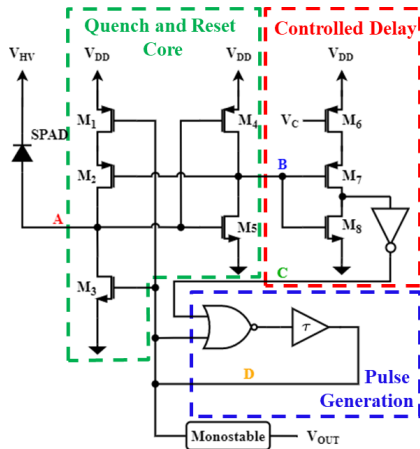


Fig. 2. Schematic of the active quench and active reset (AQR) SPAD pixel.

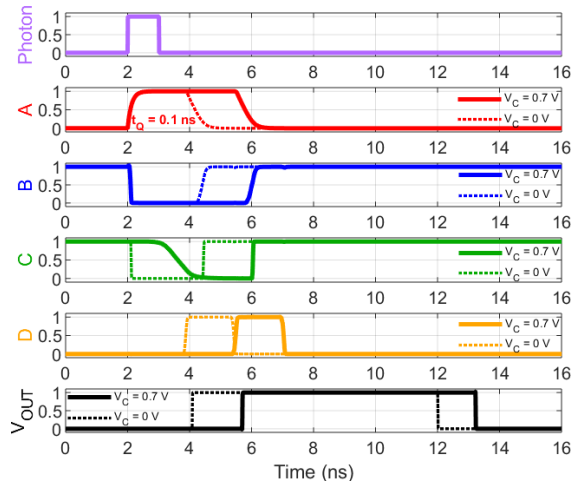


Fig. 3. Post-layout transient simulation results for the active quench and active reset (AQR) SPAD pixel.

measured to be 143Ω while the SPAD's junction capacitance was estimated to be ~ 270 fF through theoretical calculations.

The operation principle is described as follows. Initially, node A is at 0 V in a high-impedance state since the transistors M2 and M3 are off. The SPAD is biased above breakdown by the large positive voltage V_{HV} applied to the cathode. When either a photon or dark carrier initiates a self-sustaining avalanche in the SPAD, the voltage at node A will begin to increase towards $(V_{HV} - V_{BR})$ by a passive quench process through transistor M3 which provides a large resistance in the off state. The skewed inverter formed by transistors M4

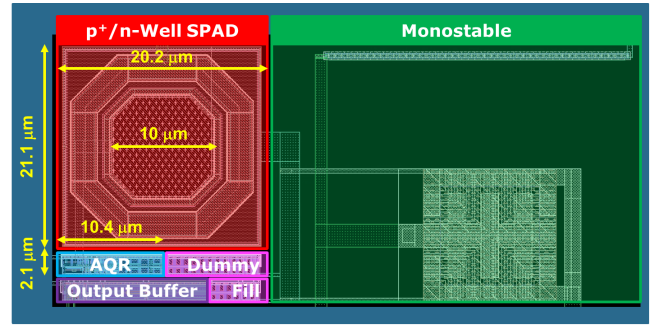


Fig. 4. Layout of SPAD pixel in a standard 65 nm CMOS process, including the output buffer and monostable.

and M5 is designed to have a low switching threshold such that the avalanche is detected early, causing the voltage at node B to fall to 0 V. As the inverter formed by M4 and M5 is cross coupled to transistor M2, then M2 turns on and creates a positive feedback path that will quickly charge node A through M1 and M2. This reduces the charge flowing through the SPAD during an avalanche, which in turn reduce the carriers trapped in the SPAD's defects which are later released to cause AP. Note that a fast positive feedback is required to achieve fast quenching. This means that the threshold voltage for the inverter formed by M4 and M5 should be optimized to a relatively low value. In our design, the threshold for this inverter is ~ 0.35 V. The fast positive feedback and the short response time of the transistors in this advanced 65 nm technology node result in this pixel achieving a very short T_Q of 0.1 ns.

Following the quench process, transistors M6 – M8 and the following inverter form a voltage-controlled delay buffer from the nodes B to C. The voltage V_C (at the gate of M6) controls the switching current of the inverter formed by M7 and M8 during its rising transition, thus adjusting the delay from nodes B to C through current starving. For example, when V_C is set to be 0 V, the PMOS M6 is fully on, and negligible delay is generated through this delay unit. This delay controls the hold-off time of the pixel, which can be increased to allow trapped charges to be released while the SPAD is biased below breakdown to limit the AP. When the falling edge of node C occurs, the following NOR gate and a 1.5 ns fixed-delay buffer are used to generate the reset pulse at node D. This reset pulse will turn off M1 and turn on M3 in order to reset node A to 0 V and restore the SPAD's excess bias. This short reset pulse would additionally be used as the SPAD's output pulse if the SPAD did not need to be directly connected off-chip, resulting in a short T_D for the SPAD pixel. However, as the measurement setup to test a single SPAD pixel required the SPAD output to drive the relatively large input capacitance of an oscilloscope probe, a monostable was used to widen the pulse widths that are sent off the chip. Without the monostable, the minimum T_D of the SPAD pixel from the post-layout simulation with V_C set to 0 V is 3.35 ns, corresponding to a maximum CR of ~ 300 MHz.

C. Layout

The layout of the AQR SPAD pixel including the monostable and output buffers is shown in Fig. 4. The size of the SPAD is $20.2 \mu\text{m} \times 20.2 \mu\text{m}$ with a $90 \mu\text{m}^2$ active area, resulting in a fill factor (FF) of 22% for the SPAD itself. The lower FF of the SPAD results from the use of a wide p-well guard ring to avoid PEB and could be improved in a further optimized implementation. The size of the AQR

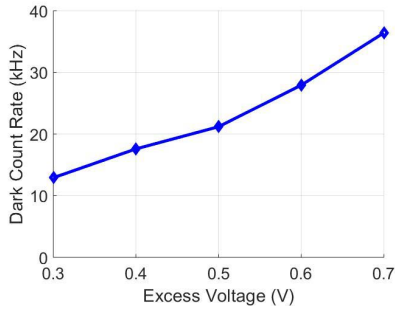


Fig. 5. Dark Count Rate (DCR) as a function of excess voltage.

front-end circuitry is only $2.1 \mu\text{m} \times 10.4 \mu\text{m}$. In a fully integrated system, the output buffer and monostable would not be required and the SPAD output would only need to drive low capacitive loads such as counters or time-to-digital converters on chip. As such, the resulting pixel pitch would be $20.2 \mu\text{m} \times 23.2 \mu\text{m}$ for an achievable FF of 19%.

III. CHARACTERIZATION AND DISCUSSION

A. Breakdown Voltage

The breakdown voltage (V_{BR}) was determined from the I-V characteristics of the SPAD using a Semiconductor Device Analyzer (Agilent B1500A). The measurements were performed on 4 different SPADs with the same structure as in the pixel design. The 4 measured results of the V_{BR} are very consistent with a small standard deviation of 17mV and an average of 9.9 V at room temperature (25 °C). This indicates that the variations of the fabrication processes are small.

B. Dark Count Rate and Afterpulsing

The inter-arrival times (IATs) of the pixel output were recorded in the dark with a digital oscilloscope (LeCroy WaveRunner 625Zi). The dark count rate (DCR) and AP can be derived from the histogram of the IAT using exponential fitting. Fig. 5 shows the DCR as a function of V_{ex} at room temperature. DCR increases with V_{ex} because enhanced electric field in the depletion region increases the avalanching probability of the dark carriers. A similar effect was observed in light-emitting devices [13]. The lifetime for the carriers trapped by the defect centers increases exponentially when temperature decreases, thus enhancing the AP probability. That is, when the temperature decreases, the AP will increase [14]. However, negligible AP was observed on our SPAD pixel even at a low temperature -35°C . Note that the V_C was set to be 0 V and negligible hold-off time was added through the voltage-controlled delay unit during the AP measurement, which indicates that the voltage-controlled delay unit is not necessary for this SPAD pixel. Transistors M6 – M8 and the following inverter can be removed in the future design. The negligible AP can be explained by the reduced avalanching charges due to the short T_Q (0.1 ns) as shown in Fig.3. Additionally, a monostable is added at the end of the signal path to widen the output pulse width so it can be measured by the oscilloscope. The measured width of output pulse is 11.5 ns, which indicates that only pulses with the IAT larger than 11.5 ns can be recorded. This could be another reason for no AP was measured in the SPAD pixel.

C. Photon Detection Probability

A Xenon lamp provides a continuous light source for the photon detection probability (PDP) measurement. Optical bandpass filters are used to select the wavelength while neutral

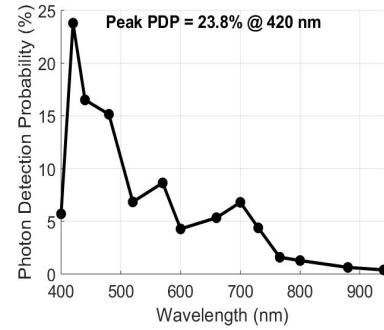


Fig. 6. Photon Detection Probability (PDP) as a function of wavelength at 0.5 V excess voltage.

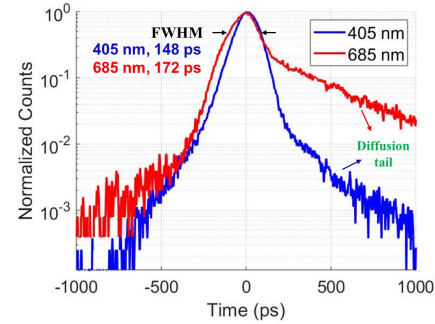


Fig. 7. Timing jitters at 0.5 V excess voltage using 405 nm and 685 nm pulsed lasers.

density filters together with the adjustment of lamp current are used to control intensity of light to ensure a single-photon PDP measurement. The PDP is defined as the ratio of the photon count rate detected by the SPAD to the incident photon rate. The photon count rate was measured using the LeCroy digital oscilloscope and the incident photon rate was calculated from the optical power measured by optical power meter (Newport 1830-C) with a calibrated silicon photodetector (Newport 818-SL). Fig. 6 shows the PDP as a function of the wavelength when the V_{ex} is 0.5 V. The peak PDP is 23.8% at the wavelength of 420 nm, in agreement with a previously published PDP measurement of a SPAD also fabricated in a standard 65 nm CMOS process [15]. The short peak wavelength at 420 nm is because of the shallow depth of p^+/n -well junction in this advanced 65 nm CMOS technology node. The fluctuation of the PDP at the longer wavelengths could be explained by the Fabry-Perot resonator formed by the stack of passivation layer and inter-metal dielectric layers in a standard CMOS process [16].

D. Timing Jitter

The timing jitter was measured using pulsed lasers with a high repetition rate. The time difference between the observed SPAD pixel output pulse and the synchronized output of a PDL 800-B laser driver was recorded by the Lecroy oscilloscope. The full-width at half-maximum (FWHM) of the time difference histogram is reported as the timing jitter. Two laser heads LDH-P-C-690 (685 nm, 70 ps FWHM) and LDH-P-C-405 (405 nm, 50 ps FWHM) were used for the timing jitter measurements. The system timing jitters are 148 ps for the 405 nm laser and 172 ps for the 685 nm laser when the V_{ex} is 0.5 V, as shown in Fig. 7. Considering the pulse width of the lasers, the timing jitters for the SPAD pixel are estimated to be 139 ps ($\sqrt{148^2 - 50^2}$) for the 405 nm laser and 157 ps

TABLE I
COMPARISON OF OUR WORK TO OTHER SPAD PIXELS WITH ACTIVE QUENCH AND RESET CIRCUITS

Year Ref.	Tech. (nm), Junction	SPAD Dia. (μm)	V _{BR} V _{ex} (V)	Quench Time (ns)	Dead Time (ns)	Max. CR (Mcps)	Peak PDP (%) @ λ (nm)	DCR (cps/μm ²)	Jitter (ps) @ λ (nm)	AP (%) @ hold-off (ns)	Circuit area (μm ²)	Simple FOM ^s
2013 [9]	350, p ⁺ /nw	20	25 6	1	20	50	-	25	-	1.3 @ 20	672	0.074
2018 [10]	350, n ⁺ /pw	80	21.7 6.6	0.48 [#]	9.5-17 [#]	105 [#]	35.1 @ 635 22 @ 850	6.1	-	4.8 @ 9.5 0.9 @ 17	17420	0.013
2019 [17]	180 HV, Customized	50	36.1 5	4	>6.2	160	-	-	-	5 @ 6.2 1.4 @ 8	32400	0.0012
2019 [18]	180 Stand., p ⁺ /nw	8	9.98 2	-	20-30	50	30.4 @ 530 3.8 @ 530	10.3k	92 @ 905	1 @ 30	400 [†]	-
2020 [11]	180 Stand., pw/DNW	10	15.5 3.5	<0.7 [#]	>5 [#]	200 [#]	34 @ 450 2.5 @ 850	20.5	-	0.75 @ 4	306	0.93
2017 [19]	65 BSI CIS /40nm ^{##} pw/DNW	15.8 *	12 4.4	-	>8	125	29.5 @ 660	4 ^{††}	75 @ 700	0.08 @ 8	25.5	-
2018 [20]	65 Stand., p ⁺ /nw	20	9.9 1.5	-	4-110	250	8 @ 470 **	2.8k	7.8	10 @ 100	540	-
This work	65 Stand., p⁺/nw	10	9.9 0.5	0.1[#]	>3.35[#]	300[#]	23.8 @ 420	233	139 @ 405 157 @ 685	~ 0	21.8	137.6

^s A simple figure-of-merit for a quench and reset circuit is defined as: FOM = Count rate [Mcps] / (Quenching time [ns] × Circuit area [μm²]). [†] Estimated from the microphoto. [#] Simulation results. ^{##} Back side illuminated 3D IC: SPAD in a 65 nm image sensing process; circuits in a 40 nm CMOS. * Estimated from the total SPAD area. ^{††} Read from the curve (maximum value). ** PDE value was reported but FF was not provided.

($\sqrt{172^2 - 70^2}$) for the 685 nm laser. As shown in Fig. 7, the diffusion tail for 685 nm laser is more apparent than from the 405 nm laser excitation. This is because photons with a shorter wavelength (405 nm) are more likely to be absorbed inside the depletion region of the shallow p⁺/n-well junction when compared to the photons with a longer wavelength (685 nm), thus showing less diffusion tail in the histogram of the timing jitter.

E. Summary and Comparison

Table I shows the summary of our work (last row) and a comparison to other SPAD pixels with AQR circuits. The technique of the positive feedback loop was used to reduce the quenching time in our AQR circuit [9]–[11]. Moreover, our pixel was fabricated in a low-cost standard 65 nm CMOS process, and it achieved the shortest quenching time and highest count rate. Additionally, the advanced technology node allows the AQC circuit to occupy the smallest area of 21.8 μm².

IV. CONCLUSION

In this work, a SPAD pixel consisting of a p⁺/n-well SPAD and a compact high-speed active quench and reset (AQR) circuit is designed and fabricated using a standard 65 nm CMOS process. The positive feedback loop in the AQR circuit allows the quenching time to be reduced to 0.1 ns. The minimum dead time of this pixel is 3.35 ns, which corresponds a count rate of ~300 Mcps. The SPAD pixel is measured to have a dark count rate of 21 kHz, a peak photon detection probability of 23.8% at 420 nm wavelength and a timing jitter of 139 ps using a 405 nm pulsed laser when it is biased at a 0.5 V excess voltage. Due to the short quenching time, negligible afterpulsing is observed even at a low temperature of −35 °C.

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